

IVVAVIK NATL PARK, YT, Canada

WMO: 719780

Lat: 69.164N

Lon: 140.150W

Elev: 800

StdP: 14.28

Time Zone: -8.00 (NAP)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -38.4	(c) -34.9	(d) -46.9	(e) 0.4	(f) -38.2	(g) -43.3	(h) 0.5	(i) -34.7	(j) 30.5	(k) 3.6	(l) 27.5	(m) 2.5	(n) 4.1	(o) 200	(p) 0.885

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.7	(c) 75.2	(d) 59.1	(e) 71.4	(f) 57.3	(g) 67.9	(h) 55.7	(i) 60.5	(j) 71.7	(k) 58.4	(l) 69.1	(m) 56.6	(n) 66.3	(o) 9.1	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
55.2	66.8	64.6	53.2	62.1	62.6	51.3	57.8	61.0	27.1	71.7	25.7	69.1	24.5	66.5	67.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 24.5	(b) 20.5	(c) 17.3	(e) -42.4	(f) 81.9	(g) 4.0	(h) 2.8	(i) -45.3	(j) 83.9	(k) -47.6	(l) 85.5	(m) -49.8	(n) 87.1	(o) -52.7	(p) 89.1		
(4) 24.5	(5) 20.5	(6) 17.3	(7) -42.6	(8) 83.5	(9) 4.0	(10) 2.1	(11) -45.5	(12) 65.0	(13) -47.8	(14) 66.2	(15) -50.0	(16) 67.4	(17) -52.9	(18) 68.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.7	-11.1	-9.1	-7.4	11.8	33.5	49.8	54.4	47.6	36.8	15.4	-1.9	-9.0
	DBStd	26.99	14.56	14.32	12.38	12.95	11.36	8.25	7.21	7.63	8.66	10.88	11.82	12.65
	HDD50	12134	1895	1653	1780	1146	519	103	40	139	401	1072	1556	1830
	HDD65	17266	2360	2073	2245	1596	976	457	334	541	846	1537	2006	2295
	CDD50	351	0	0	0	0	8	98	176	64	5	0	0	0
	CDD65	8	0	0	0	0	0	2	5	1	0	0	0	0
	CDH74	150	0	0	0	0	7	41	79	23	0	0	0	0
	CDH80	14	0	0	0	0	0	3	9	2	0	0	0	0
Wind	WSAvg	7.3	7.6	8.0	7.3	7.6	8.4	8.0	7.1	6.8	6.6	6.3	7.2	6.8
Precipitation	PrecAvg	9.8	0.5	0.4	0.4	0.3	0.5	1.2	1.8	1.8	1.1	0.9	0.6	0.5
	PrecMax	12.3	1.3	0.9	0.9	0.8	1.3	2.4	3.5	3.6	2.2	1.5	1.1	1.1
	PrecMin	7.5	0.1	0.1	0.0	0.0	0.1	0.3	0.7	0.5	0.6	0.4	0.1	0.1
	PrecStd	1.3	0.3	0.2	0.2	0.2	0.2	0.6	0.5	0.6	0.4	0.3	0.3	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.2	27.9	31.8	48.7	72.1	79.2	80.6	77.1	63.5	44.3	31.6	26.9
		MCWB	27.1	24.3	27.6	39.8	53.9	59.0	62.3	61.6	51.2	38.5	27.6	23.9
	2%	DB	22.6	23.1	23.2	42.6	64.5	73.7	75.9	70.2	58.8	37.4	25.1	19.2
		MCWB	19.8	20.6	19.9	35.9	50.2	56.9	59.9	57.3	48.6	33.4	22.2	17.0
	5%	DB	16.8	16.2	16.4	37.0	58.2	70.2	72.3	65.9	54.5	33.6	19.2	14.1
		MCWB	14.8	13.9	13.8	31.6	46.6	55.5	58.5	55.0	46.8	30.7	17.2	12.5
	10%	DB	10.2	11.4	11.3	32.0	52.3	66.4	68.8	61.4	50.4	30.0	14.5	8.5
		MCWB	8.6	9.7	9.2	28.1	43.2	53.6	57.3	53.0	44.4	27.7	12.9	7.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.2	24.6	28.0	40.0	54.9	61.2	63.9	62.6	52.1	39.4	27.8	24.0
		MCDB	30.9	27.5	31.7	47.6	71.2	74.6	77.6	75.2	61.3	43.2	31.1	26.9
	2%	WB	19.9	20.5	19.9	35.7	50.8	58.4	61.2	58.3	49.4	33.9	22.3	17.1
		MCDB	22.2	23.1	22.8	42.2	63.2	71.4	72.5	67.7	57.3	36.9	24.7	19.2
	5%	WB	14.7	14.1	13.8	31.8	47.1	56.4	59.5	55.8	47.2	31.0	17.2	12.5
		MCDB	16.8	16.2	16.2	36.7	57.8	68.3	70.1	64.1	54.0	33.5	19.1	14.3
	10%	WB	8.7	9.8	9.2	28.0	43.4	54.2	57.9	53.6	44.7	27.8	13.0	7.2
		MCDB	10.1	11.5	11.3	31.9	51.9	65.4	67.6	60.9	49.9	29.9	14.5	8.5
Mean Daily Temperature Range	5% DB	MDBR	13.3	13.8	17.6	20.1	19.1	21.6	19.7	17.5	15.3	12.9	12.4	13.2
		MCDBR	19.1	17.7	19.4	21.2	30.1	27.0	26.2	25.1	22.6	13.2	15.2	17.5
		MCWBR	16.9	15.8	16.8	15.4	17.3	13.1	13.0	13.4	10.5	13.7	16.2	
	5% WB	MCDBR	19.3	18.4	19.0	20.4	29.9	25.3	24.4	22.7	21.4	12.5	15.0	17.0
		MCWBR	17.1	16.6	16.5	14.8	17.5	12.7	12.7	12.6	12.9	10.0	13.6	15.8
Clear-Sky Solar Irradiance	taub	0.167	0.206	0.248	0.306	0.336	0.353	0.374	0.339	0.292	0.216	0.153	N/A	
	taud	1.896	2.114	2.213	2.113	2.172	2.351	2.359	2.466	2.478	2.178	1.918	N/A	
	Ebn at Noon	16	187	251	262	267	267	255	254	238	177	16	0	
	Edh at Noon	1	13	23	35	37	33	31	25	18	11	1	0	
All-Sky Solar Radiation	RadAvg	6	137	659	1399	1886	1945	1559	1043	583	189	21	0	
	RadStd	1	12	35	65	124	178	116	93	60	24	2	0	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.49	N/A	N/A	N/A	N/A	N/A	-515	-542	N/A	N/A
(47) Regional (7 neighbors)		+1.57	+1.66	N/A	N/A	N/A	N/A	-567	-583	N/A	N/A

Nomenclature: See separate page